

Design Proposal: Unbalanced Mach-Zehnder Interferometers for Extracting the Group Index of TE and TM Silicon Strip Waveguides

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Abstract — This proposal describes unbalanced Mach-Zehnder interferometers (MZIs) to be built from $500\text{ nm} \times 220\text{ nm}$ silicon strip waveguides, for both the quasi-TE and quasi-TM modes. Each polarization uses path-length differences of 50, 100 and $200\text{ }\mu\text{m}$. Two all-pass ring resonators are added for the TE mode as an independent check of the group index. The goal is to extract the waveguide group index from the measured free spectral range (FSR) and compare it with mode-solver predictions. The report covers the waveguide model, the interferometer and ring transfer functions, the derivation of the group-index extraction equation, circuit simulations, and the planned measurement analysis.

I. Introduction

The Mach-Zehnder interferometer is a basic building block of silicon photonics. It underlies modulators, switches, wavelength filters and sensors [1-3]. In all of these, device performance depends directly on how accurately we know the waveguide group index, n_g . That index sets the FSR of filters, the bandwidth of modulators, and the delay in true-time-delay lines [4].

An unbalanced MZI gives a simple, direct way to measure n_g . The FSR of the output spectrum depends only on the path-length difference ΔL and the group index. Because ΔL is defined lithographically and known precisely, a measured FSR maps directly to n_g [4].

The objectives of this design are:

- Extract n_g for the fundamental quasi-TE and quasi-TM modes of a $500\text{ nm} \times 220\text{ nm}$ strip waveguide from measured MZI spectra.
- Verify that the FSR scales as $1/\Delta L$ by using three imbalance lengths per polarization.
- Extract n_g independently from two TE ring resonators, and observe how the coupling gap changes resonance linewidth and extinction ratio.
- Compare the extracted n_g with simulation and attribute any differences to fabrication variation in waveguide width and thickness.

II. Theory

A. MZI Transfer Function

A Y-branch splits the input field E_i equally into two arms, so each arm carries $E_i/\sqrt{2}$. After propagating through arms of length L_1 and L_2 , with propagation constants β_1 and β_2 and power attenuation α , the fields reaching the combiner are:

$$E_{1,2} = \frac{E_i}{\sqrt{2}} e^{-j\beta_{1,2}L_{1,2} - \alpha L_{1,2}/2} \quad (1)$$

The Y-branch combiner sends half of each input field into the fundamental output mode; the rest is lost to radiation modes [4], [9]. The output field is therefore:

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} = \frac{E_i}{2} (e^{-j\beta_1 L_1 - \alpha L_1/2} + e^{-j\beta_2 L_2 - \alpha L_2/2}) \quad (2)$$

Both arms use the same waveguide cross-section, so $\beta_1 = \beta_2 = \beta = 2\pi n_{\text{eff}}/\lambda$. Neglecting loss, the intensity transfer function is:

$$T_{MZI}(\lambda) = \frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta \Delta L)], \quad \Delta L = L_2 - L_1 \quad (3)$$

With loss included, the arms have unequal amplitudes, which limits the extinction ratio. At 3 dB/cm, a 200 μm imbalance causes only a 0.06 dB amplitude difference between the arms, so the deep nulls predicted by (3) should still appear in the measured spectra.

B. Group Index from the Free Spectral Range

The phase difference between the two arms is:

$$\varphi(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda) \Delta L}{\lambda} \quad (4)$$

Adjacent transmission maxima occur at wavelengths λ_m and λ_{m+1} , where the phase changes by exactly 2π . The FSR is small compared with λ , so we expand φ to first order about the midpoint wavelength λ :

$$|\Delta\varphi| \approx \left| \frac{d\varphi}{d\lambda} \right| \cdot \text{FSR} = 2\pi \quad (5)$$

Differentiating (4), and remembering that n_{eff} itself depends on λ , gives

$$\frac{d\varphi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2} (n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}) = -\frac{2\pi\Delta L n_g}{\lambda^2} \quad (6)$$

The bracketed term is the definition of the group index, $n_g = n_{\text{eff}} - \lambda \cdot dn_{\text{eff}}/d\lambda$ [4]. Substituting (6) into (5):

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L} \implies n_g(\lambda) = \frac{\lambda^2}{\Delta L \cdot \text{FSR}(\lambda)} \quad (7)$$

Equation (7) is the extraction equation. The FSR is set by n_g , not n_{eff} , because waveguide dispersion shifts the fringe spacing. Each measured fringe spacing gives one value of n_g , assigned to the midpoint wavelength of that fringe pair. A single spectrum therefore yields $n_g(\lambda)$ across the measurement band.

Test with simulation data: From the INTERCONNECT spectrum of MZI2 (TE, $\Delta L = 100 \mu\text{m}$), adjacent minima occur at 1548.61 and 1554.26 nm, giving $\text{FSR} = 5.65 \text{ nm}$ at $\lambda = 1551.44 \text{ nm}$. Equation (7) then gives $n_g = (1551.44 \mu\text{m})^2 / (100 \mu\text{m} \times 5.65 \text{ nm}) = 4.26$, compared with $n_g = 4.25$ from the MODE solver, a difference of 0.2%.

C. Ring Resonator

An all-pass ring resonator consists of a ring of radius R side-coupled to a straight bus waveguide. With self-coupling coefficient r at the coupler and single-pass amplitude transmission a , the through-port transmission is [7]:

$$T_{\text{ring}}(\lambda) = \frac{a^2 - 2ra \cos \varphi + r^2}{1 - 2ra \cos \varphi + (ra)^2}, \quad \varphi = \beta L, \quad L = 2\pi R \quad (10)$$

Resonances occur when the round-trip phase φ is a multiple of 2π , producing dips in the through-port spectrum. The derivation in Section II-B applies directly with ΔL replaced by the ring circumference L :

$$FSR = \frac{\lambda^2}{n_g L} \Rightarrow n_g = \frac{\lambda^2}{2\pi R \cdot FSR} \quad (11)$$

The coupling gap sets r . A smaller gap gives stronger coupling, lower r and wider resonances. When r equals a (critical coupling), the resonance dips reach their maximum depth [7], [8]. Varying the gap at fixed radius therefore changes linewidth and extinction ratio but leaves the FSR unchanged, which isolates the effect of coupling.

III. Modelling and Simulation

A. Waveguide Geometry

The waveguide is a silicon strip, 500 nm wide and 220 nm thick, on a buried oxide layer with SiO₂ top cladding, the standard platform for silicon photonic wires [5], [11]. The fundamental quasi-TE and quasi-TM modes are both studied. The 500 nm width keeps the waveguide single-mode for each polarization at 1550 nm and matches the standard grating coupler and Y-branch library components [9-11], so no tapers are needed.

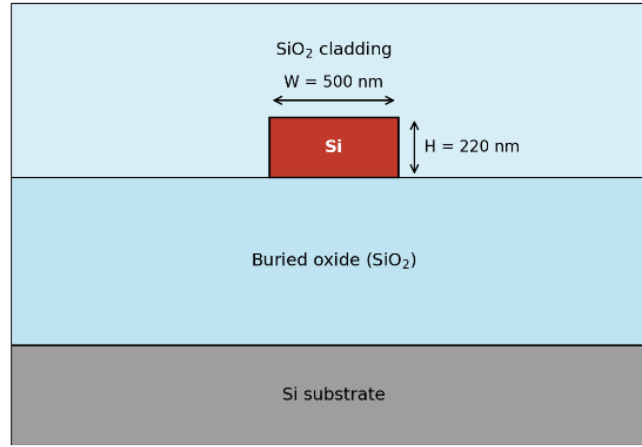


Fig. 1. Cross-section of the silicon strip waveguide.

B. Mode Profiles

The modes were computed with Lumerical MODE Solutions using the finite-difference eigenmode solver [6], [13] at $\lambda = 1550$ nm. Figs. 2 and 3 show the electric-field intensity of the quasi-TE and quasi-TM modes. The TE mode is well confined in the silicon core, with its field mostly horizontal. The TM mode extends further into the cladding above and below the core. That makes it less confined and more sensitive to the top and bottom interfaces.

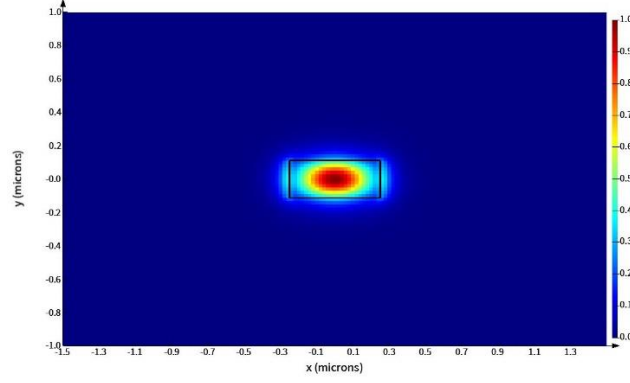


Fig. 2. Electric-field intensity of the quasi-TE mode.

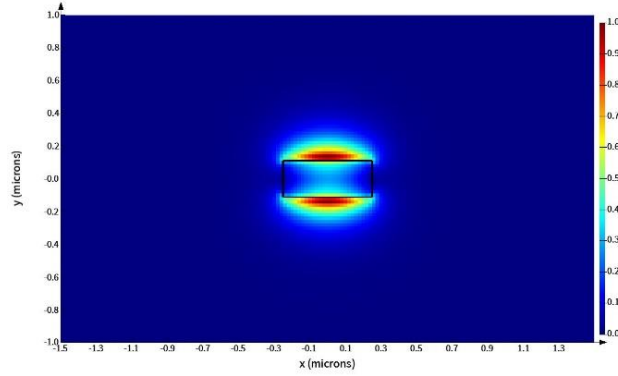


Fig. 3. Electric-field intensity of the quasi-TM mode.

C. Effective and Group Index

A frequency sweep from 1500 to 1600 nm, including material dispersion of Si and SiO₂, gives the effective and group indices in Figs. 4 and 5. Table I summarizes the values at 1550 nm. As expected, n_{eff} decreases with wavelength for both modes, because the mode spreads further into the low-index cladding at longer wavelengths. The group index is much larger than n_{eff} , which shows the strong waveguide dispersion of high-contrast silicon waveguides.

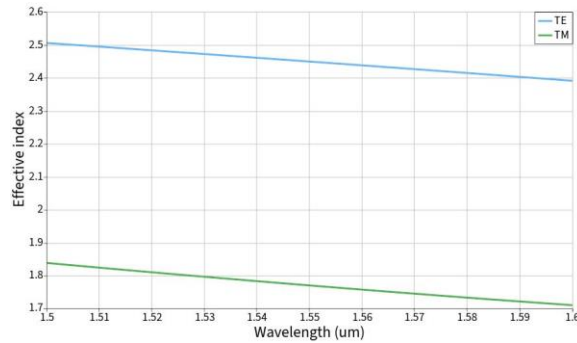


Fig. 4. Effective index versus wavelength for the quasi-TE and quasi-TM modes.

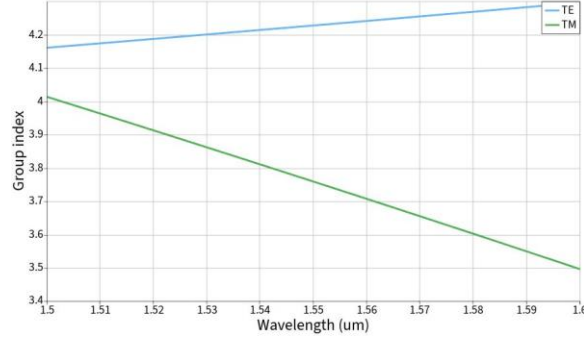


Fig. 5. Group index versus wavelength for the quasi-TE and quasi-TM modes.

TABLE I. SIMULATED MODE PROPERTIES AT 1550 NM

Mode	n_{eff}	n_g
Quasi-TE	2.45	4.25
Quasi-TM	1.77	3.77

D. Compact Model

For circuit simulation and data fitting, the effective index is described by a second-order Taylor expansion about $\lambda_0 = 1.55 \mu\text{m}$ [4]:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \quad (8)$$

Applying the group-index definition to (8) at $\lambda = \lambda_0$ gives a direct link between the compact model and the group index:

$$n_g(\lambda_0) = n_1 - \lambda_0 n_2 \quad (9)$$

The fitted coefficients (λ in μm) are listed in Table II. Using the fitted coefficients, (9) gives $n_g = 4.228$ (TE) and 3.760 (TM), within 0.5% of the solver values in Table I, which confirms the compact model.

TABLE II. COMPACT MODEL COEFFICIENTS ($\lambda_0 = 1.55 \mu\text{m}$)

Mode	n_1	$n_2 (\mu\text{m}^{-1})$	$n_3 (\mu\text{m}^{-2})$
Quasi-TE	2.4500	-1.1472	-0.4376
Quasi-TM	1.7706	-1.2832	1.6689

E. Design Variations

Table III lists the eight devices. The ΔL values were chosen so that every design fits several fringes in the $\sim 50 \text{ nm}$ measurement bandwidth. The smallest imbalance, $50 \mu\text{m}$, gives about four fringes. The largest, $200 \mu\text{m}$, gives more than fifteen, which improves the averaging of n_g . Expected FSRs were calculated from (7) at 1550 nm , using $n_g = 4.25$ for TE and 3.77 for TM from Table I.

TABLE III. DESIGN VARIATIONS AND EXPECTED FSR AT 1550 NM

Device	Polarization	ΔL or ring (μm)	Expected FSR (nm)
MZI1	TE	50	11.31

Device	Polarization	ΔL or ring (μm)	Expected FSR (nm)
MZI2	TE	100	5.65
MZI3	TE	200	2.83
MZI4	TM	50	12.75
MZI5	TM	100	6.37
MZI6	TM	200	3.19
Ring1	TE	R = 10, gap 0.20	9.00
Ring2	TE	R = 10, gap 0.30	9.00

The FSR also follows the expected $1/\Delta L$ trend: halving ΔL doubles the FSR. For the rings, $L = 2\pi R = 62.8 \mu\text{m}$, which gives about five resonances in the measurement band. The two rings share the same radius, so their FSRs should match; only the gap differs. All devices use 500 nm strip waveguides with no tapers. TE devices use $10 \mu\text{m}$ bend radii. TM devices use larger bend radii, because the TM mode is less confined and has higher bend loss at small radii [4].

F. Circuit Simulation

The circuits were modelled in Lumerical INTERCONNECT [13] (Fig. 6). Each circuit has an input grating coupler, a Y-branch splitter, two waveguide arms, a Y-branch combiner and an output grating coupler. The waveguide arms use the compact models from Section III-D. The grating couplers (TE and TM versions) and Y-branches use measured S-parameter models from the SiEPIC EBeam library [9-11], [14]. The waveguide loss is set to 3 dB/cm, a typical value for strip waveguides in this process [4]. The same Y-branch model (ebeam_y_1550) was used for both polarizations; a back-to-back test with TM grating couplers confirmed that its S-parameter data includes the TM mode.

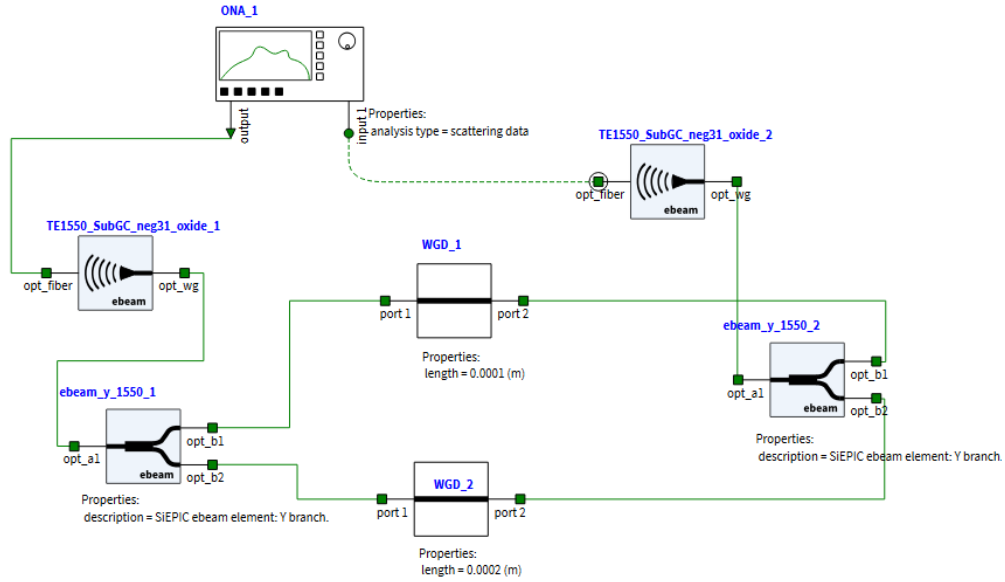


Fig. 6. MZI circuit model in Lumerical INTERCONNECT (TE, $\Delta L = 100 \mu\text{m}$). The TM circuit uses the same layout with TM grating couplers.

Figs. 7 and 8 show the simulated spectra. The fringes follow the grating coupler envelope, which peaks near 1550 nm and limits the usable bandwidth. The simulated FSRs near 1550 nm are 11.33, 5.65 and 2.80 nm for $\Delta L = 50, 100$ and $200 \mu\text{m}$, which agree with the values in Table III to within 1.1%. For the TM MZIs, the simulated FSRs near 1550 nm are 13.0, 6.0 and 3.0 nm for $\Delta L = 50, 100$ and $200 \mu\text{m}$, which agree with the values in Table III to within 6%.

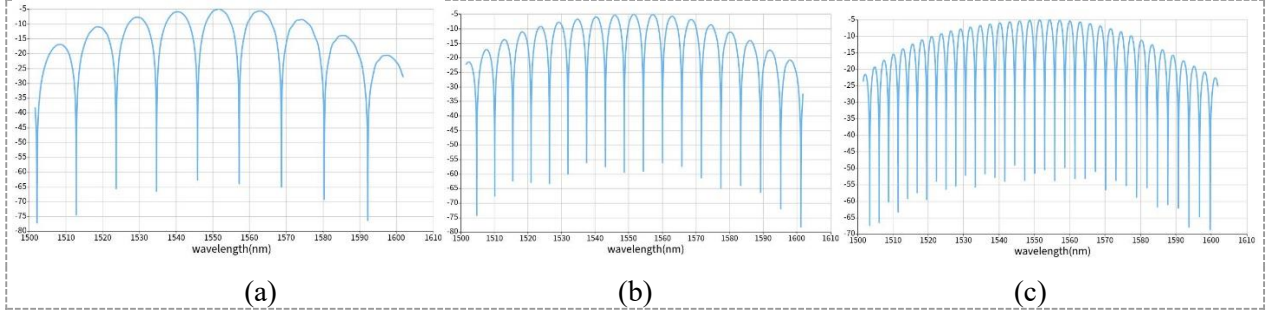


Fig. 7. Simulated transmission spectra of the TE MZIs: (a) $\Delta L = 50 \mu\text{m}$ (FSR = 11.33 nm), (b) $\Delta L = 100 \mu\text{m}$ (FSR = 5.65 nm), (c) $\Delta L = 200 \mu\text{m}$ (FSR = 2.80 nm).

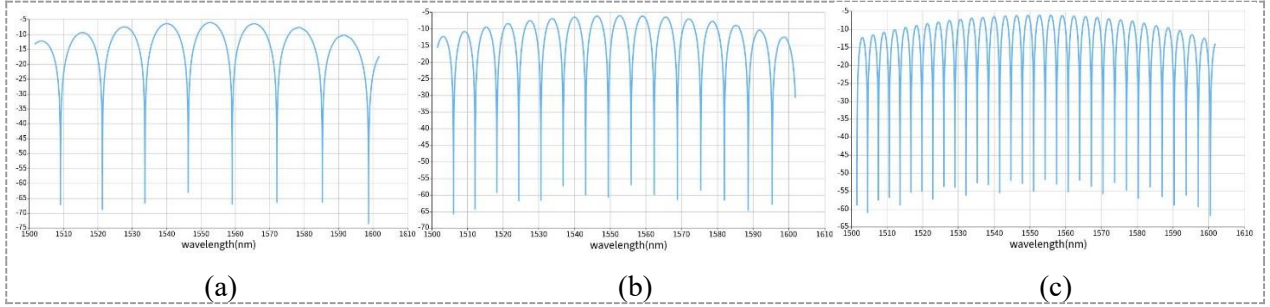


Fig. 8. Simulated transmission spectra of the TM MZIs: (a) $\Delta L = 50 \mu\text{m}$ (FSR = 13.0 nm), (b) $\Delta L = 100 \mu\text{m}$ (FSR = 6.0 nm), (c) $\Delta L = 200 \mu\text{m}$ (FSR = 3.0 nm).

The ring resonators will be modelled with the directional-coupler half-ring cell from the component library [14], using the same TE grating couplers and a straight waveguide of length πR to close the ring. Based on (9) and the TE group index in Table I, both rings are expected to show an FSR of about 9.0 nm near 1550 nm. Ring2 (gap 300 nm) should show narrower resonances than Ring1 (gap 200 nm) because of its weaker coupling. These circuit simulations are in progress and will be added in the next revision.

G. Planned Analysis of Measured Data

The measured spectra will be processed in four steps:

- Normalize the grating coupler response [10] by fitting a low-order polynomial to the spectral envelope and subtracting it in dB.
- Locate the transmission nulls. Nulls are sharper than maxima, so their positions can be found more precisely.
- Compute the FSR between each pair of adjacent nulls and apply (7) to get n_g at each fringe midpoint, giving $n_g(\lambda)$ across the band.
- As a check, fit the full transfer function (3) to the normalized spectrum using the compact model (10) with n_1 , n_2 and n_3 as free parameters, then get n_g from (11).

For the rings, the resonance dips are located the same way, and (9) gives n_g from the spacing between adjacent resonances. Fitting each dip with (8) also gives r and a , which shows how the coupling gap affects

the resonance. Because each polarization has three ΔL values, n_g can be extracted three independent ways from the MZIs, plus a fourth way from the rings for TE. Agreement between them confirms the analysis. Any consistent offset from the simulated n_g most likely reflects fabricated width or thickness differing from the nominal $500 \text{ nm} \times 220 \text{ nm}$. The TM mode is expected to be more sensitive to thickness, and the TE mode to width. Wafer-scale studies of this kind of variation [12] will guide the interpretation.

IV. Fabrication

[To be completed]

V. Experimental Data

[To be completed.]

VI. Analysis

[To be completed.]

VII. Conclusion

[To be completed.]

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